INP - [redacted] Angelo dos Santos - Exercícios 2.2.
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6 $Y = g(X) \sim Uniforme(0,2)$

PIT: $F_X(X) \sim Uniforme(0,1)$

$F_X(x) = \int_0^x 2(1-t)\,dt$

$= \left[2\left(t - \frac{t^2}{2}\right)\right]_0^x =$

$\left[2t - t^2\right]_0^x = 2x - x^2$

$(2x - x^2) \sim Uniforme(0,1) \Rightarrow (4x - 2x^2) \sim Uniforme(0,2)$

$\therefore Y = 4x - 2x^2 = g(X)$ □

14 Amplitude $= X_{(n)} - X_{(1)}$

$X \sim Exp(2) \Rightarrow F_X(x) = \int_0^x 2e^{-2t}\,dt = 1 - e^{-2x}, \quad x \geq 0$

$P(X_{(1)} > \mu, X_{(n)} \leq v) = P(\mu < X_i < v, \forall i \in \{1,2,\dots,n\})$

$\overset{ind}{=} P(\mu < X_1 < v) \cdot P(\mu < X_2 < v) \cdots P(\mu < X_n < v) \overset{i.d.}{=} [P(\mu < X < v)]^n$

$= [F_X(v) - F_X(\mu)]^n$

$\frac{\partial}{\partial v}[F_X(v) - F_X(\mu)]^n = n[F_X(v) - F_X(\mu)]^{n-1} \cdot f_X(v)$

$-\frac{\partial}{\partial \mu}[n(F_X(v) - F_X(\mu))^{n-1} f_X(v)] = (n-1)(n)(F_X(v) - F_X(\mu))^{n-2} \cdot f_X(v) \cdot f_X(\mu)$

$= n(n-1)[F_X(v) - F_X(\mu)]^{n-2} f_X(\mu) f_X(v)$

$P(R \leq r) = \int_0^{\infty} \int_{\mu}^{\mu+r} n(n-1)[F_X(v) - F_X(\mu)]^{n-2} f_X(v) f_X(\mu)\,dv\,d\mu$

$\int_{\mu}^{\mu+r} (n-1)[F_X(\mu) - F_X(v)]^{n-2} f_X(v)\,dv = \left[(F_X(v) - F_X(\mu))^{n-1}\right]_{\mu}^{\mu+r}$

$= [F_X(\mu+r) - F_X(\mu)]^{n-1}$

$\hookrightarrow (1 - e^{-2(\mu+r)}) - (1 - e^{-2\mu}) = e^{-2\mu} \cdot e^{-2\mu} \cdot e^{-2r}$

$= e^{-2\mu} \cdot (1 - e^{-2r})$

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14 (Continuação)

$$P(R \leq r) = n(1-e^{-2r})^{n-1} \int_0^\infty e^{-2(n-1)u} \cdot 2e^{-2u}\, du$$

$$P(R \leq r) = n(1-e^{-2r})^{n-1} \int_0^\infty 2e^{-2nu}\, du$$

Dado que $\int_0^\infty 2e^{-2nu}\, du = \left[\frac{-e^{-2nu}}{n}\right]_0^\infty = \frac{1}{n}$

$$P(R < r) = n(1-e^{-2r})^{n-1}\left(\frac{1}{n}\right) = (1-e^{-2r})^{n-1}$$

$r = 4 \Rightarrow P(R \leq 4) = (1-e^{-2(4)})^{n-1} = (1-e^{-8})^{n-1}$ □

24 $E[R] = E[X_{(3)} - X_{(1)}] = E[X_{(3)}] - E[X_{(1)}]$

$X \sim$ Uniforme $(0,1) \Rightarrow F_X(x) = x,\ 0 < x < 1$

A k-ésima estatística de ordem $X_{(k)}$ é menor ou igual a um valor x se, e somente se, pelo menos k das n observações amostrais forem menos ou iguais a x: $P(X_{(k)} \leq x) = P(\text{número de } X_i \leq x \text{ é} \geq k)$

$Y \sim$ Binomial $(n, F_X(x))$

$$\Rightarrow F_{X_{(k)}}(x) = P(Y \geq k) = \sum_{j=k}^{n} \binom{n}{j} [F_X(x)]^j [1-F_X(x)]^{n-j}$$

$$f_{X_{(k)}}(x) = \frac{d}{dx} \sum_{j=k}^{n} \binom{n}{j} [F_X(x)]^j [1-F_X(x)]^{n-j}$$

$$\frac{d}{dx}\left([F_X(x)]^j [1-F_X(x)]^{n-j}\right) = j[F_X(x)]^{j-1} f_X(x)[1-F_X(x)]^{n-j} - (n-j)[F_X(x)]^j [1-F_X(x)]^{n-j-1} f_X(x)$$

$$\binom{n}{j} j = n\binom{n-1}{j-1}\ ;\ \binom{n}{j}(n-j) = n\binom{n-1}{j}$$

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24 (Continuação)

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!\,(n-k)!}\,[F_X(x)]^{k-1}\,[1-F_X(x)]^{n-k}\,f_X(x)$$

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!\,(n-k)!}\,x^{k-1}\,(1-x)^{n-k}, \quad 0<x<1$$

$\therefore X_{(k)} \sim \text{Beta}(k, n-k+1)$

$$E[X_{(k)}] = \frac{k}{n+1} \quad ; \quad \overset{n=3}{(k=1)} \Rightarrow E[X_{(1)}] = \frac{1}{3+1} = \frac{1}{4}$$

$$E[X_{(3)}] = \frac{3}{4} \Rightarrow E[R] = \frac{3}{4} - \frac{1}{4} = \frac{1}{2} \qquad \square$$

25 $X \sim \text{Uniforme}(0,1)$

Logo $F_X(x) = x, \quad 0<x<1$

$V = X_{(n)} = \max(X_1, X_2, \ldots, X_n)$

$$F_V(v) = P(V \le v) = P(X_1 \le v, X_2 \le v, \ldots, X_n \le v) \overset{ind}{=} \prod_{i=1}^{n} P(X_i \le v)$$

$$\overset{i.d.}{=} [F_X(v)]^n \Rightarrow f_V(v) = \frac{d}{dv}[F_X(v)]^n = n\,[F_X(v)]^{n-1} \cdot f_X(v)$$

Para $u<v$:

$$P(u < X_i < v) = F_{X_i}(v) - F_{X_i}(u) \overset{i.d.}{=} [F_X(v) - F_X(u)]^n$$

$$F_{U,V}(u,v) = P(U \le u, V \le v) = P(V \le v) - P(U > u, V \le v)$$

$$= [F_X(v)]^n - [F_X(v) - F_X(u)]^n$$

$$f_{U,V}(u,v) = \frac{\partial^2}{\partial u\,\partial v} F_{U,V}(u,v)$$

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25 (Continuação)

$$\frac{\partial}{\partial u} F_{U,V}(u,v) = -n[F(v)-F(u)]^{n-1} \cdot (-f(u)) = n[F(v)-F(u)]^{n-1} f(u)$$

$$f_{U,V}(u,v) = n(n-1)[F(v)-F(u)]^{n-2} f(u) f(v), \text{ para } u<v$$

$f(x)=1$ e $F(x)=x$; $x \in (0,1)$:

$$f_{U,V}(u,v) = n(n-1)(v-u)^{n-2}, \quad 0<u<v<1$$

$$\begin{cases} R = V-U \\ U = U \end{cases} \Rightarrow \begin{cases} V = U+R \\ U = U \end{cases}$$

$$\det(J) = \begin{vmatrix} \frac{\partial u}{\partial u} & \frac{\partial u}{\partial r} \\ \frac{\partial v}{\partial u} & \frac{\partial v}{\partial r} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1$$

Como $0<u<v<1$, as novas variáveis satisfazem $0<r<1$ e $0<v<1-r$

$$f_{U,R}(u,r) = n(n-1)r^{n-2}, \quad 0<u<1-r, \quad 0<r<1.$$

A marginal de R:

$$f_R(r) = \int_0^{1-r} n(n-1)r^{n-2}\,du = n(n-1)r^{n-2}(1-r), \quad 0<r<1$$

$R \sim \text{Beta}(n-1, 2)$

$n=3 \Rightarrow R \sim \text{Beta}(2,2)$

$$\text{Var}(R) = \frac{2.2}{(2+2)^2(2+2+1)} = \frac{4}{4^2 \cdot 5} = \frac{1}{20} \quad \square$$